White Wine Quality Intelligence & Prediction System

Dataset: UCI White Wine (N = 4,898) **Framework:** Streamlit + Plotly (Option B) **Selected Model:** Random Forest Classifier

1. Business Problem & Project Objectives

SPECIFICATION 1

BUSINESS PROBLEM CONTEXT

Traditional Sensory Evaluation: Evaluating wine quality in the commercial industry relies on human tasting panels.

- **High Cost & Slow Speed:** Organizing human tasting panels incurs high recurring labor and scheduling costs.
- **Subjectivity & Bias:** Panel ratings suffer from human fatigue, personal palate bias, and environmental inconsistencies.
- **Lack of Predictive Control:** Winemakers lack an automated mechanism to verify quality before bottling.

KEY PROJECT OBJECTIVES

1. **Chemical Feature Analysis:** Quantify the impact of physicochemical properties (alcohol, acidity, density) on wine ratings.
2. **Supervised Predictive Modeling:** Train and compare regression and classification models to forecast quality categories.
3. **Interactive Dashboard Deployment:** Build a real-time web interface in Streamlit for decision support and live inference.

2. Dataset Overview & Attributes

SPECIFICATION 2

DATASET ORIGIN & SCOPE

- **Source:** UCI Machine Learning Repository (Portuguese "Vinho Verde" White Wine dataset).
- **Sample Size:** `N = 4,898` total observations.
- **Target Variable:** Continuous sensory quality score rated on a 0 to 10 scale (actual observed values span 3 to 9).
- **Average Score:** `5.87 / 10` baseline continuous mean.

11 CONTINUOUS PHYSICOCHEMICAL FEATURES

CHEMICAL FEATURE	ROLE / DESCRIPTION
Alcohol (%)	Percent alcohol by volume in wine sample
Volatile Acidity (g/L)	Acetic acid amount (vinegar taste at high levels)
Free & Total SO_2 (mg/L)	Sulfur dioxide prevents microbial growth & oxidation
Density (g/cm³) & Residual Sugar	Determines mouthfeel, body weight & sweetness
Fixed Acidity, pH, Chlorides, Citric Acid	Tartaric acid profile, saltiness & crispness balance

3. Data Cleaning & Preprocessing

SPECIFICATION 3

DATA HYGIENE STEPS

- **Missing Value Audit:** Confirmed 0 missing/null cells across all 4,898 rows and 12 columns.
- **Deduplication:** Identified and removed duplicate observations to eliminate data leakage across cross-validation splits.
- **Outlier Treatment:** Inspected high values in residual sugar and sulfur dioxide using Interquartile Range (IQR) bounds. Valid chemical extremes were retained.

TARGET BINARIZATION & STRATIFICATION

- **Binary Target Definition:**
  - **Good Wine (Class 1):** Rating ≥ 7 (~21.6% of dataset).
  - **Not Good Wine (Class 0):** Rating < 7 (~78.4% of dataset).
- **Class Imbalance Strategy:** Stratified 80/20 train/test splitting was implemented to maintain exact class proportions during validation.

4. Exploratory Data Analysis & Correlations

SPECIFICATION 4

KEY CORRELATION RESULTS (PEARSON R)

- **Alcohol vs. Quality:** Strongest positive association (`r = +0.44`). Higher alcohol content directly aligns with higher sensory ratings.
- **Volatile Acidity vs. Quality:** Primary negative association (`r = -0.19`). Elevated volatile acidity causes off-flavors.
- **Density Interactions:** Density exhibits strong correlation with residual sugar (`r = +0.84`) and inverse correlation with alcohol (`r = -0.68`).

DISTRIBUTION ANALYSIS

Target Distribution: Continuous quality ratings follow a normal-like curve concentrated at ratings 5 and 6, with extreme ratings (3, 4, 8, 9) being rare.

Statistical Takeaway: Linear correlations exist, but chemical interactions (e.g., alcohol vs. sugar balance) are non-linear, necessitating flexible modeling.

5. Ordinary Least Squares (OLS) Regression

SPECIFICATION 5

OLS MODEL FORMULATION

The OLS model attempts to model continuous quality as a linear combination of features:

$$\text{Quality} = \beta_0 + \beta_1(\text{Alcohol}) + \beta_2(\text{Volatile Acidity}) + \ldots + \varepsilon$$

- **Linearity Assumption:** Assumes each chemical feature exerts a constant, additive effect on wine rating.
- **Multicollinearity Sensitivity:** OLS coefficients suffer from instability when collinear variables (density & sugar) are included simultaneously.

OLS PERFORMANCE METRICS

- **R^2 Score:** `0.291` (Explains less than 30% of total variance).
- **Mean Absolute Error (MAE):** `0.58` rating points.
- **Root Mean Squared Error (RMSE):** `0.74` rating points.
- **Conclusion:** OLS underfits the data due to strong non-linear chemical interactions in winemaking.

6. Model Comparison & Evaluation Metrics

SPECIFICATIONS 6 & 7

REGRESSION VS. CLASSIFICATION EVALUATION SUMMARY

MODEL ARCHITECTURE	TASK FRAME	ACCURACY / R^2	PRECISION (GOOD)	RECALL (GOOD)	F1-SCORE	ROC-AUC
Linear Regression (OLS)	Regression	$R^2 = 0.291$	N/A	N/A	N/A	N/A
Logistic Regression	Classification	78.2%	0.520	0.280	0.360	0.751
Decision Tree Classifier	Classification	76.5%	0.480	0.450	0.464	0.724
Random Forest Classifier	**Classification**	**84.5%**	**0.710**	**0.580**	**0.638**	**0.862**

MODEL SELECTION RATIONALE

The **Random Forest Classifier** is selected as the winning architecture because it achieves the highest ROC-AUC (`0.862`) and F1-Score (`0.638`), effectively overcoming class imbalance and non-linear decision boundaries.

7. Permutation Feature Importance

SPECIFICATION 7

TOP PREDICTIVE VARIABLES (PERMUTATION IMPORTANCE)

1. **Alcohol (%):** Importance Mean = **0.342** (Primary positive predictor).
2. **Volatile Acidity (g/L):** Importance Mean = **0.221** (Primary negative predictor).
3. **Free Sulfur Dioxide (mg/L):** Importance Mean = **0.125** (Key preservative).
4. **Density (g/cm³):** Importance Mean = **0.094** (Body indicator).
5. **Residual Sugar (g/L):** Importance Mean = **0.062** (Sweetness balance).

METHODOLOGY & VALUE

Why Permutation Importance over Gini Importance?

Permutation importance shuffles feature values on unseen test data to observe the drop in ROC-AUC score. Unlike Gini impurity, it avoids favoring continuous features with high cardinality and accurately reflects actual model generalization power.

8. Interactive Streamlit Dashboard (`app.py`)

SPECIFICATION 8

DASHBOARD ARCHITECTURE (OPTION B: STREAMLIT + PLOTLY)

- **Summary KPI Cards:** High-visibility displays of Total Observations (4,898) and Average Quality Score (5.87 / 10).
- **Interactive Controls & Filters:** Sidebar sliders for Alcohol, Volatile Acidity, and Density, plus model view dropdowns.
- **Tabbed Interface:**
  - EDA & Distributions (Pie chart & Correlation Heatmap)
  - Model Performance & Regression Actual-vs-Predicted Plots
  - Classification Confusion Matrix (`px.imshow`)

REAL-TIME MODEL INFERENCE ENGINE

Winemakers can dynamically manipulate input sliders in the sidebar. The application evaluates the inputs against the trained model logic and displays:

- **Instant Classification:** "Good Wine" vs. "Not Good Wine".
- **Probability Gauge Chart:** Displays confidence score percentage in real time.

9. Limitations, Future Work & Defense Q&A

LIMITATIONS & FUTURE ENHANCEMENTS

Project Limitations:

- Sensory panel ratings introduce human subjectivity into target labels.
- Lacks vineyard metadata (grape variety, vintage year, region, price point).

Future Enhancements:

- Evaluate Gradient Boosting algorithms (XGBoost, LightGBM).
- Add automated CSV batch scoring to the Streamlit application.

DEFENSE Q&A SCRIPT (SPECIFICATION 10)

Q1: Why classification over regression?

A: Ratings cluster around 5–6 due to human tasting habits. Binary classification provides actionable commercial quality control.

Q2: Why did Random Forest win?

A: OLS assumes straight additive relationships, while Random Forest captures complex non-linear chemical thresholds.